

Lie Groups and Lie Algebras

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1 Matrix Exponential and Log

Definition 1.1. Recall that we define the exponential function by

$$e^x = \exp(x) := \sum_{k=0}^{\infty} \frac{1}{k!} x^k$$

Definition 1.2. We can define the matrix exponential in a similar way with $1 = I_n$. Let $X \in M_{n \times n}(\mathbb{C})$ then we define the matrix exponential by

$$\exp(X) = I + X + \frac{X^2}{2!} + \cdots$$

Examples.

(1) Let

$$X = \begin{pmatrix} 0 & -t \\ t & 0 \end{pmatrix}$$

note that

$$X^2 = -t^2 I_2, \quad X^3 = -t^2 X, \quad X^4 = t^4 I_2, \dots$$

then

$$\begin{aligned} \exp(X) &= I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots \\ &= \begin{pmatrix} 1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \cdots & -t + \frac{t^3}{3!} - \frac{t^5}{5!} + \cdots \\ t - \frac{t^3}{3!} + \frac{t^5}{5!} - \cdots & 1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \cdots \end{pmatrix} \\ &= \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} \end{aligned}$$

where is exactly the matrix for rotation by t anticlockwise.

(2) Let

$$A = \begin{pmatrix} 0 & x \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{pmatrix}$$

observe that A, B are in fact nilpotent matrix.

Definition 1.3 (Matrix norm). A matrix norm is any function $\|\cdot\| : M_{n \times n}(\mathbb{R}) \rightarrow \mathbb{R}$ satisfying the following conditions

- (i) $\|A + B\| \leq \|A\| + \|B\|$ for all $A, B \in M_{n \times n}(\mathbb{R})$.
- (ii) $\|\lambda A\| = |\lambda| \|A\|$ for all $\lambda \in \mathbb{R}$ and $A \in M_{n \times n}(\mathbb{R})$.
- (iii) $\|A\| = 0$ if and only if $A = 0$.

Definition 1.4 (Operator norm). The operator norm is defined by

$$\|X\|_{op} = \sup\{|Xv| \mid v \in \mathbb{R}^n, |v| \leq 1\}$$

where $|v| = \sqrt{x_1^2 + \cdots + x_n^2}$. Note that $|v| \leq 1$ is the unit ball and $|Xv|$ is a solid ellipsoid. Unless specified, we will use $\|\cdot\|$ to represent $\|\cdot\|_{op}$ in the following notes.

Lemma 1.5. For any $v \in \mathbb{R}^n$, $X, Y \in M_{n \times n}(\mathbb{R})$ we have

- (1) $|Xv| \leq \|X\| |v|$.
- (2) $\|XY\| \leq \|X\| \|Y\|$.
- (3) $\|X^m\| \leq \|X\|^m$.

Proof.

(1) Suppose $|v| = 1$, then the result is trivial by definition. Now suppose $|v| > 1$ then

$$|Xv| = |v| \left| X \frac{v}{|v|} \right| \leq |v| \|X\| \leq \|X\| |v|$$

(2) Suppose $|v| \leq 1$ then by (1) we have

$$|XYv| \leq \|X\| |Yv| \leq \|X\| \|Y\| |v| \leq \|X\| \|Y\|$$

(3) This is basically the repeated applications of (2).

□

Definition 1.6. A sequence (X_k) of matrices converges to a matrix X if $\forall \varepsilon > 0, \exists N$ such that

$$\|X_k - X\| < \varepsilon \quad \forall k \geq N$$

Theorem 1.7 (Weierstrass M -test). If $f_k(X)$ is a sequence of functions such that

$$\|f_k(X)\| \leq M_k \quad \forall k, X$$

then if $\sum_{k \geq 0} M_k$ converges, then $\sum_{k \geq 0} f_k(X)$ converges uniformly and absolutely to a function $f(X)$.

Proof. See Analysis 4. □

Remark. Remember $\exp$ doesn't converge uniformly everywhere (only on a bounded set) i.e. $\|X\| \leq R$ where R is the radius of convergence.

Theorem 1.8. $\exp X$ converges absolutely for any square matrix X .

Proof. Let $\|X\| < R$ where R is the radius of convergence and let $f_n(X) = \frac{X^n}{n!}$, consider

$$\|f_k(X)\| = \left\| \frac{X^k}{k!} \right\| \leq \frac{\|X\|^k}{k!} \leq \frac{R^k}{k!}$$

Let

$$M_k := \frac{R^k}{k!}$$

then

$$\sum_{k \geq 0} M_k = \sum_{k \geq 0} \frac{R^k}{k!} \rightarrow \exp R \text{ as } k \rightarrow \infty$$

Then by Weierstrass M -test we then know that $\exp X$ converges absolutely. □

Proposition 1.9. The matrix exponential has the following properties

(1)

$$\frac{d}{dt}(\exp tX) = X \exp(tX).$$

(2) If $XY = YX$, then

$$\exp X \exp Y = \exp(X + Y).$$

(3) $\exp X$ is invertible with inverse $\exp(-X)$.

Proof.

(1) By definition

$$\begin{aligned} \frac{d}{dt}(\exp tX) &= \frac{d}{dt} \left(\sum_{k \geq 0} \frac{(tX)^k}{k!} \right) \\ &\stackrel{\text{uniform convergence}}{=} \sum_{k \geq 0} \frac{d}{dt} \left(\frac{(tX)^k}{k!} \right) \\ &= \sum_{k \geq 1} \frac{k}{k!} t^{k-1} X^k = \sum_{k \geq 1} \frac{1}{(k-1)!} t^{k-1} X^k \\ &= X \sum_{k \geq 1} \frac{(tX)^{k-1}}{(k-1)!} = X \exp(tX) \end{aligned}$$

(2) Note that

$$\begin{aligned} \exp X \exp Y &= \left(\sum_{i \geq 0} \frac{X^i}{i!} \right) \left(\sum_{j \geq 0} \frac{Y^j}{j!} \right) \\ &= \sum_{k \geq 0} \sum_{i=0}^k \frac{1}{i!(k-i)!} X^i Y^{k-i} \\ &= \sum_{k \geq 0} \frac{(X + Y)^k}{k!} = \exp(X + Y) \end{aligned}$$

since $XY = YX$.

(3) Let $Y = -X$ in (2) immediately gives the result. □

Remark. In fact, in general

$$\frac{d}{dt} \exp(X + tY) \neq Y \exp(X + tY).$$

Remark. We have shown that $\exp : \text{all } n \times n \text{ matrix} \rightarrow \text{invertible } n \times n \text{ matrices}$ i.e.

$$\exp : \mathfrak{gl}(n, \mathbb{R}) \rightarrow GL(n, \mathbb{R})$$

Definition 1.10. Suppose $F : \mathbb{R}^N \rightarrow \mathbb{R}^N$ then we can write

$$F = \begin{pmatrix} F_1(x_1, \dots, x_N) \\ \vdots \\ F_N(x_1, \dots, x_N) \end{pmatrix}$$

We define $d_p F$ to be the matrix of partial derivatives at $p \in U \subset \mathbb{R}^N$ open such that

$$d_p F = \begin{pmatrix} \frac{\partial F_1}{\partial x_1}(p) & \dots & \frac{\partial F_1}{\partial x_N}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial F_N}{\partial x_1}(p) & \dots & \frac{\partial F_N}{\partial x_N}(p) \end{pmatrix}$$

We think of this as a linear map that is the best approximation to F at p in the sense that $F(p+v) = F(p) + d_p F v + o(|v|)$. Another way of thinking of $d_p F(v)$ is a directional derivative of F at p in the v direction

$$d_p F(v) = \left. \frac{d}{dt} \right|_{t=0} F(p + tv)$$

Theorem 1.11 (Inverse Function Theorem). Suppose $F : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is continuously differentiable. If $d_p F$ is invertible, then there exists $p \in U \subset \mathbb{R}^N$ open and $F(p) \in V \subset \mathbb{R}^N$ open such that $F(U) = V$ and $F|_U : U \rightarrow V$ is a bijection. Moreover, F^{-1} is differentiable with derivative $d_{F(p)} F^{-1} = (d_p F)^{-1}$.

Theorem 1.12 (Local invertibility of $\exp$). There exists an open neighbourhood $U \subset \mathfrak{gl}(n, \mathbb{R})$ containing 0 and $V \subset GL(n, \mathbb{R})$ of I_n such that $\exp(U) = V$ and $\exp|_U : U \rightarrow V$ is bijective. Then we call the inverse $\log V \rightarrow U$.

Proof. We can re-imagine $\mathfrak{gl}(n, \mathbb{R})$ as $\mathbb{R}^{n^2}$ and same $GL(n, \mathbb{R})$ so we have

$$\exp : \mathbb{R}^{n^2} \rightarrow \mathbb{R}^{n^2}$$

We need to check that $d_0 \exp$ is bijective. It's enough to check the case for $n = 2$ where $\exp : \mathfrak{gl}(2, \mathbb{R}) \rightarrow GL(2, \mathbb{R})$ and

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

then $\exp A \in GL(2, \mathbb{R})$ and

$$\exp A = \begin{pmatrix} (\exp A)_{11}(A_{11}, A_{12}, A_{21}, A_{22}) \\ (\exp A)_{12}(A_{11}, A_{12}, A_{21}, A_{22}) \\ (\exp A)_{21}(A_{11}, A_{12}, A_{21}, A_{22}) \\ (\exp A)_{22}(A_{11}, A_{12}, A_{21}, A_{22}) \end{pmatrix}$$

Then

$$d_A \exp = \begin{pmatrix} \frac{\partial(\exp A)_{11}}{\partial A_{11}} & \dots & \frac{\partial(\exp A)_{11}}{\partial A_{22}} \\ \vdots & \ddots & \vdots \\ \frac{\partial(\exp A)_{22}}{\partial A_{11}} & \dots & \frac{\partial(\exp A)_{22}}{\partial A_{22}} \end{pmatrix}$$

Evaluating at $A_{11}, A_{12}, A_{21}, A_{22} = 0$ and by Taylor expansion

$$\exp(A) = I + A + \frac{A^2}{2} + \dots$$

we deduce that $d_0 \exp A = A$ i.e. $d_0 \exp = I_n$ which is invertible. □

Remark. $\log$ is infinitely differentiable and even analytic so it admits Taylor expansion close to I_n

$$\log(I + M) = M - \frac{1}{2}M^2 + \frac{1}{3}M^3 - \frac{1}{4}M^4 + \dots$$

for M small, so that $I + M$ is closed to I .

Theorem 1.13 (Baker-Campbell-Hausdorff (BCH)). The BCH formula tells us what $\log(\exp X \exp Y)$ is even when X, Y don't commute. It tells us this for X, Y close to 0 so that $\exp X$ and $\exp Y$ are closed to I_n :

$$\log(\exp X \exp Y) = X + Y + \frac{1}{2}[X, Y] + h.o.t.$$

and we denote $[X, Y] = XY - YX$ the commutator or Lie bracket. In particular, if X, Y commute then $\exp X \exp Y = \exp(X + Y)$ which implies $\log(\exp X \exp Y) = X + Y$. If X, Y don't commute we can use Taylor series expansion.

Proof. Close to 0 we have

$$\begin{aligned}\exp X \exp Y &= (I + X + \frac{X^2}{2} + o(\varepsilon^2))(I + Y + \frac{Y^2}{2} + o(\varepsilon^2)) \\ &= I + X + Y + \underbrace{\frac{X^2}{2} + \frac{Y^2}{2} + XY}_{M} + o(\varepsilon^2)\end{aligned}$$

then

$$\begin{aligned}\log(I + M) &= M - \frac{1}{2}M^2 + \frac{1}{3}M^3 - \frac{1}{4}M^4 + \dots \\ &= X + Y + \frac{X^2}{2} + \frac{Y^2}{2} + XY - \frac{1}{2}(X^2 + Y^2 + XY + YX) + o(\varepsilon^2) \\ &= X + Y + \frac{1}{2}(XY - YX) + o(\varepsilon^2)\end{aligned}$$

□

Remark.

- (1) There are infinitely many higher order terms.
- (2) You can write a formula for the nth order term (Dykin).
- (3) All higher order terms consist of iterated commutator brackets.
- (4) The group multiplication on $GL(n, \mathbb{R})$ is determined by the Lie bracket of $\mathfrak{gl}(n, \mathbb{R})$.

2 Lie Groups

Definition 2.1. A matrix Lie group is a subgroup G of $GL(n, \mathbb{R})$ such that G is topologically closed in $GL(n, \mathbb{R})$.

Definition 2.2. A subgroup G of $GL(n, \mathbb{R})$ is topologically closed if for any sequence $g_1, g_2, \dots$ of elements of G we have $g_n \rightarrow g \in G$.

Remark.

- (1) Matrix Lie groups are instances of more general object called Lie group.
- (2) Lie groups are groups which admit local coordinate systems like $\mathbb{R}^n$ for which the group multiplication and the map $g \rightarrow g^{-1}$ are smooth.

To prove that G is a Lie group we need to check

- (i) G is a subgroup of $GL(n, \mathbb{R})$.
- (ii) G is topologically closed.

Example. Let $SL(n, \mathbb{R})$ defined by

$$SL(n, \mathbb{R}) = \{g \in GL(n, \mathbb{R}) \mid \det g = 1\}$$

is a matrix Lie group.

Example. The orthogonal group $O(n)$ defined by

$$O(n) = \{g \in GL(n, \mathbb{R}) \mid gg^T = I\}$$

In particular

$$SO(n) = \{g \in O(n) \mid \det(g) = 1\}$$

Proposition 2.3. $O(n)$ is a matrix Lie group.

Proof. We first need to show $O(n)$ is a subgroup of $GL(n, \mathbb{R})$. Pick $g, h \in O(n)$ and we want to show that $gh^{-1} \in O(n)$. Consider $(gh^{-1})(gh^{-1})^T = gh^{-1}(h^{-1})^T g^T = gg^T = I$ which implies $gh^{-1} \in O(n)$. Thus, $O(n)$ is a subgroup of $GL(n, \mathbb{R})$.

Now we need to show that $O(n)$ is closed in $GL(n, \mathbb{R})$. If we define $\varphi : g \mapsto gg^T$ then $\varphi^{-1}(\{I\})$ is closed since I is closed and φ is obviously continuous. $\square$

Remark. Since $SO(n) = O(n) \cap SL(n, \mathbb{R})$ and we know that the intersection of two closed sets are closed and the intersection of two subsets is also a subset. Thus $SO(n)$ is also a matrix Lie group.

Proposition 2.4. Let G, H be matrix Lie groups in $GL(n, \mathbb{R})$. Then $G \cap H$ is a matrix Lie group.

Proof. See Remark above. $\square$

Example. Unitary Groups defined by

$$U(n) = \{g \in GL(n, \mathbb{C}) \mid gg^\dagger = I_n\}$$

similarly we have

$$SU(n) = \{g \in U(n) \mid \det g = 1\}$$

are a matrix Lie groups.

3 Lie algebras

What have we done so far?

$$\exp : \mathfrak{gl}(n, \mathbb{R}) \rightarrow GL(n, \mathbb{R})$$

which is locally invertible i.e. there are neighbourhood u of $0 \in \mathfrak{gl}(n, \mathbb{R})$ and V of $I_n \in GL(n, \mathbb{R})$ such that $\log : V \rightarrow U$ is the inverse of $\exp$.

We also have BCH where

$$\exp X \exp Y = \exp C$$

where C is some expression involving Lie brackets on $\mathfrak{gl}(n, \mathbb{R})$.

Our goal is to construct a replacement of $\mathfrak{gl}(n, \mathbb{R})$, whenever we replace $GL(n, \mathbb{R})$ with a matrix Lie group G such that

$$\exp : \mathfrak{g} \rightarrow G$$

will satisfy the properties we talked about. We want to have a Lie bracket on $\mathfrak{g}$.

Example. Think about $\{z \in \mathbb{C} \mid |z| = 1\} = U(1)$ which is a matrix Lie group. Then if $z \in U(1)$, we can write $z = \exp(i\theta)$ where $\theta \in \mathbb{R}$. Then

$$\begin{aligned} \exp : \mathfrak{g} &\rightarrow G \\ i\theta &\mapsto \exp(i\theta) \end{aligned}$$

note that $i\theta$ is exactly the tangent of the circle at the identity.

Theorem 3.1. Let G be a matrix Lie group defined by

$$\text{Lie}(G) = \mathfrak{g} = \{X \in \mathfrak{gl}(n, \mathbb{R}) \mid \exp(tX) \in G \text{ for all } t \in \mathbb{R}\}$$

Then

- (i) $\mathfrak{g}$ is a vector space.
- (ii) $X, Y \in \mathfrak{g}$ implies $[X, Y] \in \mathfrak{g}$.
- (iii) $\mathfrak{g}$ is parallel to the tangent space at identity in G .
- (iv) $\exp : \mathfrak{g} \rightarrow G$ is locally invertible.

We call $\mathfrak{g}$ the Lie algebra of G .

Example. $O(n) = \{g \in GL(n, \mathbb{R}) \mid g^T g = I\}$. Geometrically, g can be reimagined to be transformations that preserve the dot product. We can think of $O(n)$ as rotations and reflections. And

$$\text{Lie}(O(n)) = o(n) = \{X \in \mathfrak{gl}(n, \mathbb{R}) \mid \exp(tX) \in O(n), \text{ for all } t \in \mathbb{R}\}$$

And $\exp(tX) \in O(n) \implies (\exp(tX))^T (\exp(tX)) = I_n$. Thus

$$\exp(tX^T) \exp(tX) = I_n \implies X^T = -X$$

In this case

$$o(n) = \{X \in \mathfrak{gl}(n, \mathbb{R}) \mid X^T + X = 0\}$$

To check $o(n)$ is a Lie algebra we need to show $o(n)$ is preserved under $[\cdot, \cdot]$ i.e. for any $X, Y \in o(n)$, $[X, Y] \in o(n)$ and this is left as an exercise.

Example. Consider $SL(2, \mathbb{C})$ defined by

$$SL(2, \mathbb{C}) = \{g \in GL(2, \mathbb{C}) \mid \det(g) = 1\}$$

The Lie algebra of $SL(2, \mathbb{C})$ can be defined by

$$sl(2, \mathbb{C}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid \det \left(\exp \left(t \begin{pmatrix} a & b \\ c & d \end{pmatrix} \right) \right) = 1 \text{ for all } t \in \mathbb{R} \right\}$$

Let's consider $\det \left(\exp \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} \right) \right)$ to first order in t then

$$\begin{aligned} \det \left(\begin{pmatrix} 1+at & bt \\ ct & 1+dt \end{pmatrix} + O(t^2) \right) &= (1+at)(1+dt) - bct^2 + O(t^2) = 1 + (a+d)t + O(t^2) = 1 \\ &\implies a+d=0 \end{aligned}$$

thus

$$sl(2, \mathbb{C}) = \left\{ \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \mid a, b, c \in \mathbb{C} \right\}$$

notice this is a 3-dimensional vector space with the basis

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Note that

$$[H, X] = 2X, \quad [X, Y] = H, \quad [H, Y] = -2Y$$

Definition 3.2. Let $X \subseteq \mathbb{R}^n$, $\underline{v} \in X$. A differentiable path through $\underline{v}$ in X is a differentiable function

$$\gamma : \mathbb{R} \rightarrow X \text{ such that } \gamma(0) = \underline{v}$$

Theorem 3.3. If $\gamma(s)$ is a path in G such that $\gamma(0) = I$ then $\frac{d\gamma}{ds}(0) \in \mathfrak{g}$.

Proof. We need to prove $\exp\left(t \frac{d\gamma}{ds}(0)\right) \in G$ for all $t \in \mathbb{R}$. It is sufficient to prove that $\exp\left(\frac{d\gamma}{ds}(0)\right) \in G$. Define $h(s) = \log \gamma(s)$ for s sufficiently close to 0, so that $\gamma(s)$ is close to I_n . We then have

$$\begin{aligned} \frac{dh}{ds}(0) &= (d_1 \log)\left(\frac{d\gamma}{ds}(0)\right) \\ &= (d_0 \exp)\left(\frac{d\gamma}{ds}(0)\right) \\ &= id\left(\frac{d\gamma}{ds}(0)\right) = \frac{d\gamma}{ds}(0) \end{aligned}$$

So it is sufficient to prove $\exp\left(\frac{dh}{ds}(0)\right) \in G$.

$$\frac{dh}{ds}(0) = \lim_{\varepsilon \rightarrow 0} \frac{h(\varepsilon) - h(0)}{\varepsilon} = \lim_{n \rightarrow \infty} nh\left(\frac{1}{n}\right)$$

We need to show

$$\exp\left(nh\left(\frac{1}{n}\right)\right) = \exp\left(n \log \gamma\left(\frac{1}{n}\right)\right) = \gamma^n\left(\frac{1}{n}\right)$$

Since $\gamma(s) \in G$ for all s and since G is a group we have

$$\gamma^n(s) \in G, \forall s \implies \gamma^n\left(\frac{1}{n}\right) \in G$$

□

Example. Take $\gamma(s) = \begin{pmatrix} \cos s & -\sin s \\ \sin s & \cos s \end{pmatrix} \in O(2)$. This is a path of rotation matrices then

$$\frac{d\gamma}{ds} = \begin{pmatrix} -\sin s & -\cos s \\ \cos s & -\sin s \end{pmatrix}$$

Evaluating at $s = 0$ we have $\frac{d\gamma}{ds}(0) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ which is exactly a anti-symmetric matrix so it belongs to the Lie algebra.

Definition 3.4. If $\gamma(s)$ is a path in $\mathbb{R}^n$ then $\frac{d\gamma}{ds}$ is called the tangent vector field along γ . Then the theorem tells us that $\mathfrak{g}$ contains all tangent vectors to paths in G passing through I_n . We will call this the tangent space of G at the identity. This is often denoted by $T_I G$.

Lemma 3.5. If $X \in \mathfrak{g}$ there exists $\gamma(s) \in G$ such that $\frac{d\gamma}{ds}(0) = X$ and $\gamma(0) = I$.

Proof. Take $\gamma(s) = \exp sX$.

□

Theorem 3.6. Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a differentiable function and let

$$A = \{v \in \mathbb{R}^n \mid F(v) = c, c \in \mathbb{R}^m\}$$

If $v \in A$ and $D_v F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is surjective then $T_v A = \ker(D_v F)$.

Example. Let $G = O(n) = \{g \in \mathfrak{gl}(n, \mathbb{R}) \mid gg^T = I_n\}$. Define

$$\begin{aligned} F : \mathfrak{gl}(n, \mathbb{R}) &\rightarrow S \\ g &\mapsto gg^T \end{aligned}$$

where S is the set of symmetric matrices.

Observe that $(gg^T)^T = gg^T$, hence $F(g) \in S$. To compute $D_I F$ consider

$$\begin{aligned} F(I + \varepsilon X) &= (I + \varepsilon X)(I + \varepsilon X)^T \\ &= (I + \varepsilon X)(I + \varepsilon X^T) \\ &= I + \varepsilon(X + X^T) + O(\varepsilon^2) \end{aligned}$$

So $D_I F : \mathfrak{gl}(n, \mathbb{R}) \rightarrow S$ is given by $X \mapsto X + X^T$. To show that $D_I F$ is surjective let $Y = X + X^T$ symmetric, then for X symmetric we have $X = X^T$, hence we can take $X = 1/2Y$. Thus

$$\mathfrak{o}(n) = \ker(D_I F) = \{X \in \mathfrak{gl}(n, \mathbb{R}) \mid X + X^T = 0\}$$

which is the set of skew symmetric matrices.

Example. Let $G = U(n) = \{g \in \mathfrak{gl}(n, \mathbb{C}) \mid gg^\dagger = I_n\}$. Define

$$F : \mathfrak{gl}(n, \mathbb{C}) \rightarrow H$$

$$g \mapsto gg^\dagger$$

where H is the set of Hermitian matrices.

Observe that $(gg^\dagger)^\dagger = (\bar{g}g^T)^T = gg^\dagger$, hence $F(g) \in H$. To compute $D_I F$ consider

$$\begin{aligned} F(I + \varepsilon X) &= (I + \varepsilon X)(I + \varepsilon X)^\dagger \\ &= (I + \varepsilon X)(I + \varepsilon X^\dagger) \\ &= I + \varepsilon(X + X^\dagger) + O(\varepsilon^2) \end{aligned}$$

So $D_I F : \mathfrak{gl}(n, \mathbb{C}) \rightarrow H$ is given by $X \rightarrow X + X^\dagger$. Similarly as before $D_I F$ is surjective. Then we must have

$$u(n) = \ker(D_I F) = \{X \in \mathfrak{gl}(n, \mathbb{C}) \mid X + X^\dagger = 0\}$$

which is the set of skew Hermitian matrices.

Definition 3.7. A Lie algebra is a vector space $\mathfrak{g}$ equipped with an operation

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$$

satisfying

(i) Bilinearity

$$\begin{aligned} [aX + bY, Z] &= a[X, Z] + b[Y, Z] \\ [X, aY + bZ] &= a[X, Y] + b[X, Z] \end{aligned}$$

for all $X, Y, Z \in \mathfrak{g}$ and $a, b \in \mathbb{F}$.

(ii) For all $X \in \mathfrak{g}$ we require $[X, X] = 0$.

(iii) The Jacobi identity holds for all $X, Y, Z \in \mathfrak{g}$

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$$

Remark. $[X, X] = 0$ implies $[X, Y] = [-Y, X]$. The converse is also true, provided we are working over a field of char not equal to 2.

Theorem 3.8. Any finitely dimensional Lie algebra over a field $\mathbb{F}$ is isomorphic to a Lie subalgebra of $\mathfrak{gl}(n, \mathbb{F})$ for

Theorem 3.9. Let G be a matrix Lie group. Define

$$\text{Lie}(G) = \mathfrak{g} = \{X \in \mathfrak{gl}(n, \mathbb{R}) \mid \exp tX \in G \text{ for all } t \in \mathbb{R}\}$$

Then

(i) $\mathfrak{g}$ is a vector space.

(ii) If $X, Y \in \mathfrak{g}$ then $[X, Y] \in \mathfrak{g}$.

(iii) $\mathfrak{g}$ is parallel to tangent space of G at I .

(iv) $\exp : \mathfrak{g} \rightarrow G$ is locally invertible.

Proof. It is enough to prove (ii) i.e. if $X, Y \in \mathfrak{g}$ then $[X, Y] \in \mathfrak{g}$. To show that something lives in $\mathfrak{g}$ we need to construct a path

$$\gamma(s) \in G \text{ such that } \gamma(0) = I \text{ and } \frac{d\gamma}{ds}(0) = [X, Y]$$

Try $\gamma(s) = \exp(X\sqrt{s}) \exp(Y\sqrt{s}) \exp(-X\sqrt{s}) \exp(-Y\sqrt{s})$. Notice that $\gamma(0) = I$. We will evaluate $\gamma(s)$ using BCH. Consider first

$$\exp(X\sqrt{s}) \exp(Y\sqrt{s}) = \exp\left((X + Y)\sqrt{s} + \frac{s}{2}[X, Y] + O(s^{3/2})\right)$$

and also

$$\exp(-X\sqrt{s}) \exp(-Y\sqrt{s}) = \exp\left(-(X + Y)\sqrt{s} + \frac{s}{2}[X, Y] + O(s^{3/2})\right)$$

Multiplying the two expressions using BCH we get

$$\begin{aligned} \gamma(s) &= ((X + Y)\sqrt{s} - (X + Y)\sqrt{s} + s[X, Y] - s[X + Y, X + Y] + O(s^{3/2})) \\ &= \exp\left(s[X, Y] + O(s^{3/2})\right) \end{aligned}$$

So

$$\begin{aligned} \frac{d\gamma}{ds} &= ([X, Y] + O(s^{1/2})) \exp\left(s[X, Y] + O(s^{3/2})\right) \\ \frac{d\gamma}{ds}(0) &= [X, Y] \end{aligned}$$

□

4 Homomorphisms

Definition 4.1. Recall a homomorphism $F : (G, *) \rightarrow (H, \cdot)$ is a map such that

$$F(g_1 * g_2) = F(g_1) \cdot F(g_2) \text{ and } F(1_G) = 1_H$$

for all $g_1, g_2 \in G$.

Consider

$$\det : GL(n, \mathbb{R}) \rightarrow GL(1, \mathbb{R})$$

$$A \mapsto \det A$$

is a homomorphism. Recall $\exp : \mathfrak{gl}(n, \mathbb{R}) \rightarrow GL(n, \mathbb{R})$ is not invertible in general but is locally invertible i.e. $\exists U \subset \mathfrak{gl}(n, \mathbb{R}), 0 \in U, V \subset GL(n, \mathbb{R})$ such that $I \in V$ then $\exp : U \rightarrow V$ is invertible. We can think of this as providing local coordinates near $I \in GL(n, \mathbb{R})$, namely

$$\exp \left(\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \right)$$

gives a parametrisation of V . So anything in V can be written in this form. We can think of a_{ij} as local coordinates i.e. via the exp map local coordinates on $\mathfrak{g}$ near 0 give us local coordinates near I in G we call this exponential chart.

Definition 4.2. If G is a matrix Lie group with Lie algebra $\mathfrak{g}$ and $0 \in U \subset \mathfrak{g}$ and $I \in V \subset G$ are neighbourhoods such that $\exp : U \rightarrow V$ is a bijection then we call $\exp : U \rightarrow V$ the exponential chart.

Definition 4.3. A homomorphism $F : G \rightarrow H$ of matrix Lie group is smooth if it is smooth when written in local exponential charts. This means the following diagram commutes:

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{f} & \mathfrak{h} \\ \downarrow \exp & \exp \downarrow \nearrow \log & \\ G & \xrightarrow{F} & H \end{array}$$

We would like to get a map $f : \mathfrak{g} \rightarrow \mathfrak{h}$ as $\mathfrak{g}, \mathfrak{h}$ are vector spaces we can think of them as some $\mathbb{R}^n$.

Remark. $F(\exp X) = \exp(f(X))$ for all X in a neighbourhood of $0 \in \mathfrak{g}$,

Theorem 4.4. Let $F : G \rightarrow H$ be a smooth homomorphism of matrix groups. Then

- (1) f is the restriction of a linear map $F_* : \mathfrak{g} \rightarrow \mathfrak{h}$ to a neighbourhood of 0.
- (2) $F(\exp X) = \exp(f(X))$ for all $X \in \mathfrak{g}$.
- (3) $F_*[X, Y] = [F_*X, F_*Y]$ for all $X, Y \in \mathfrak{g}$.

Example. We have seen that

$$\det : GL(n, \mathbb{R}) \rightarrow GL(1, \mathbb{R})$$

$$A \mapsto \det A$$

is a group homomorphism which means there exists a linear map

$$\det_* : \mathfrak{gl}(n, \mathbb{R}) \rightarrow \mathfrak{gl}(n, \mathbb{R})$$

such that

$$\det(\exp A) = \exp \left(\det_* (A) \right) \text{ for all } A \in \mathfrak{gl}(n, \mathbb{R})$$

where Trace is exactly what we want.

Definition 4.5. A one-parameter subgroup of a matrix group G is a smooth homomorphism

$$F : (\mathbb{R}, +) \rightarrow G$$

Example. For any $X \in \mathfrak{g}$, $F(t) = \exp(tX)$ is a one-parameter subgroup.

Lemma 4.6. Any one-parameter subgroup $G : \mathbb{R} \rightarrow G$ has this form i.e. $\exists X \in \mathfrak{g}$ such that $F(t) = \exp(tX)$.

Proof. As we want F to be a one-parameter subgroup, we want

$$F(s+t) = F(s)F(t) \text{ and } F(0) = I$$

We will differentiate the first equation to get a differential equation

$$\begin{aligned}\frac{d}{ds}\big|_{s=0} F(s+t) &= F'(t) \\ \frac{d}{ds}\big|_{s=0} F(s)F(t) &= F'(0)F(t)\end{aligned}$$

We can check $F(t) = \exp(tF'(0))$ is a solution. □

Proof of Theorem 4.4. Notice that for all $X \in \mathfrak{g}$ $\exp tX$ is a one-parameter subgroup in G . Since F is a smooth homomorphism, and since one-parameter subgroup is just a smooth homomorphism from $\mathbb{R}$ then $F(\exp tX)$ is a one-parameter subgroup in H . Hence

$$F(\exp tX) = \exp tY \text{ for some } Y \in \mathfrak{h}$$

For some t $\exp tX$ and $\exp tY$ are both closed to the identity so live in the domain of the local log. Hence

$$\begin{aligned}F(\exp tX) &= \exp f(tX), \text{ for small } t \\ F(\exp tX) &= \exp(tY) \\ \implies \exp f(tX) &= \exp tY\end{aligned}$$

And for small t we can take log such that

$$f(tX) = tY = td_0f(X) + h.o.t.$$

The second line follows from the Taylor expansion of the smooth map f . Hence $d_0f(X) = Y$ and the higher order terms vanish. This implies

$$f(tX) = tY = td_0f(X) = d_0f(tX)$$

since d_0f is a linear map. We conclude that $f = d_0f$ on exponential charts. So $F_* = d_0f$, where d_0f is the matrix with entries $(d_0f)_{ij} = \frac{\partial f_i}{\partial x_j}$ which proves (1).

So take $t = 1$ and $F(\exp X) = \exp(f(X))$ for all $X \in \mathfrak{g}$.

To prove (3), i.e. F_* preserves the commutator consider

$$\exp tX \exp tY \exp(-tX) \exp(-tY)$$

and use BCH we get

$$\begin{aligned}\exp tX \exp tY &= \exp\left(t(X+Y) + \frac{t^2}{2}[X, Y] + O(t^3)\right) \\ \exp(-tX) \exp(-tY) &= \exp\left(-t(X+Y) + \frac{t^2}{2}[X, Y] + O(t^3)\right) \\ \implies \exp tX \exp tY \exp(-tX) \exp(-tY) &= \exp(t^2[X, Y] + O(t^3))\end{aligned}$$

Applying F to the whole expression we get

$$F(\exp(t^2[X, Y] + O(t^3))) = \exp(t^2F_*[X, Y] + O(t^3))$$

Also

$$\begin{aligned}F(\exp tX \exp tY \exp(-tX) \exp(-tY)) &= F(\exp tX)F(\exp tY)F(\exp(-tX))F(\exp(-tY)) \\ &= \exp(tF_*X) \exp(tF_*Y) \exp(-tF_*X) \exp(-tF_*Y) \\ &= \exp(t^2[F_*X, F_*Y] + O(t^3))\end{aligned}$$

If we take t to be small and take logs we have

$$t^2[F_*X, F_*Y] = t^2F_*[X, Y] + O(t^3)$$

Differentiate twice with respect to t and let $t = 0$ we get

$$[F_*X, F_*Y] = F_*[X, Y]$$

□

5 Connected and simply connected groups

Definition 5.1. Recall a path from x to y in X is a continuous map $\gamma : [0, 1] \rightarrow X$ such that $\gamma(0) = x$ and $\gamma(1) = y$.

Definition 5.2. A set X is path-connected if any two points in X can be joined by a path.

Proposition 5.3. Let G be path-connected matrix Lie group then if $U \subset G$ neighbourhood of I then U generates G .

Proof. Choose $g \in G$, let γ be the path that connects I to g i.e. $\gamma : [0, 1] \rightarrow G$ such that $\gamma(0) = I$ and $\gamma(1) = g$. Now choose $\varepsilon > 0$ as in the Lemma below and N such that $\frac{1}{N} < \varepsilon$ so we can partition $[0, 1]$ into N bits.

$$I = \gamma(0), \gamma\left(\frac{1}{N}\right), \dots, \gamma\left(\frac{N}{N}\right) = g$$

Hence

$$\gamma\left(\frac{i}{N}\right)^{-1} \gamma\left(\frac{i+1}{N}\right) \in U$$

since $\left|\frac{i+1}{N} - \frac{i}{N}\right| = \left|\frac{1}{N}\right| < \varepsilon$. Thus

$$\gamma\left(\frac{i+1}{N}\right) = \gamma\left(\frac{i}{N}\right) u_{i+1}, \quad u_{i+1} \in U$$

Then

$$\gamma\left(\frac{1}{N}\right) = I u_1 = u_1 \in U$$

$$\gamma\left(\frac{2}{N}\right) = u_1 u_2$$

$\vdots$

$$\gamma\left(\frac{N}{N}\right) = u_1 \cdots u_N \in U$$

□

Lemma 5.4. Let $\gamma : [0, 1] \rightarrow G$ be continuous path, $U \subset G$ neighbourhood of I then there exists $\varepsilon > 0$ such that

$$|t - s| < \varepsilon \implies \gamma(t)^{-1} \gamma(s) \in U$$

Proof. Suppose there exists some sequence $t_n, s_n \in [0, 1]$ such that $|t_n - s_n| \rightarrow 0$ but $\gamma(t_n)^{-1} \gamma(s_n) \notin U$. By passing to a subsequence of t_n and using Bolzano-Weierstrass we can deduce $|t_n - t| \rightarrow 0$. By the triangle inequality we can show $s_n \rightarrow t$. But γ is continuous i.e.

$$\gamma(t_n)^{-1} \gamma(s_n) \rightarrow \gamma(t)^{-1} \gamma(t) = I$$

Since for large n , $\gamma(t_n)^{-1} \gamma(s_n) \in U$ which gives a contradiction. □

Corollary 5.5. Let $F : G \rightarrow H$ be smooth homomorphism. If G is path-connected then F is determined by $F_* : \mathfrak{g} \rightarrow \mathfrak{h}$.

Proof. Assume G is path-connected. Notice that for $u' \in U \subset G$, we have $\log(u) \in \mathfrak{g}$ so $F(u) = \exp(F_*(\log(u)))$. Now choose $g \in G$, by the Proposition we know that $g = u_1 \cdots u_N$ for $u_i \in U$ and since F is homomorphism

$$F(g) = F(u_1) \cdots F(u_N) = \exp(F_*(\log(u_1))) \cdots \exp(F_*(\log(u_N)))$$

□

Definition 5.6. A set X is simply connected if for all $a, b \in X$ there is a path γ from a to b and if we have two paths γ, δ from a to b in X , then there is a homotopy $h(x, t)$ from $\gamma(x)$ to $\delta(x)$.

Definition 5.7. A homotopy between $\gamma(x)$ and $\delta(x)$ is a continuous function $h : [0, 1] \times [0, 1] \rightarrow X$ such that $h(x, 0) = \gamma(x)$, $h(x, 1) = \delta(x)$. And $h(0, t) = \gamma(0) = \delta(0)$, $h(1, t) = \gamma(1) = \delta(1)$.

Definition 5.8. X is simply connected if for every loop $\gamma \in X$, there exists null-homotopy of γ .

Theorem 5.9. Let G, H be matrix Lie groups, G is simply connected. Then for every Lie algebra homomorphism $f : \mathfrak{g} \rightarrow \mathfrak{h}$, there is a unique smooth homomorphism $F : G \rightarrow H$ such that $F_* = D_I F = f$.

Proof. We already have a local homomorphism

$$\begin{array}{ccc}
\mathfrak{g} & \xrightarrow{f} & \mathfrak{h} \\
\downarrow \exp & & \exp \downarrow \tilde{\gamma} \log \\
U \subset G & \xrightarrow{F_0} & H
\end{array}$$

$F_0 : U \rightarrow H$ and $F_0(u) = \exp(f(\log u))$. We wish to extend F_0 from U to the whole of G . Since G is simply connected, it is path-connected. We can take a path $\gamma : [0, 1] \rightarrow G$ such that $\gamma(0) = I$ and $\gamma(1) = g \in G$. If we take a partition

$$0 = x_0 < x_1 < \cdots < x_N = 1$$

such that $|x_{i+1} - x_i| < \varepsilon$ given $\varepsilon > 0$. Then we must have $\gamma^{-1}(x_i)\gamma(x_{i+1}) \in U$. In particular, $g = \gamma(1) = \gamma(x_N) = \gamma(x_0)^{-1}\gamma(x_1)\gamma(x_1)^{-1}\cdots\gamma(x_{N+1})^{-1}\gamma(x_N)$. In other word, $g = u_1u_2\cdots u_N$ for $u_1, \dots, u_N \in U$. Therefore,

$$F(g) = F_0(u_1) \cdots F_0(u_N)$$

We shall check first that $F(g)$ doesn't depend on the partition. Let y be a partition between x_i and x_{i+1} . Then

$$g = \gamma(x_0)^{-1}\gamma(x_1)\cdots\gamma(x_i)^{-1}\gamma(y)\gamma(y)^{-1}\gamma(x_{i+1})\cdots$$

Then we replace $F_0(u_i)$ by $F_0(v)$ and $F_0(w)$. Notice that since $y \in (x_i, x_{i+1})$ we know that $|x_{i+1} - x_i| < \varepsilon$ then

$$|x_i - y| < \varepsilon \text{ and } |y - x_{i+1}| < \varepsilon$$

i.e. $v, w \in U$. Moreover, use the fact that F_0 is a homomorphism so $F_0(u_i) = F_0(v)F_0(w)$.

Now we show that $F(g)$ doesn't depend on the path. Suppose we have two paths $\gamma, \delta : [0, 1] \rightarrow G$ such that $\gamma(0) = \delta(0) = I$ and $\gamma(1) = \delta(1) = g$. Since G is simply connected, there exists function $h : [0, 1] \times [0, 1] \rightarrow G$ such that

$$h(x, 0) = \gamma(x), \quad h(x, 1) = \delta(x), \quad h(0, t) = I, \quad h(1, t) = g$$

We are going to use a result that there exists $\varepsilon > 0$ such that if $|x - y| < \varepsilon$ and $|t - s| < \varepsilon$ we have $h(x, t)^{-1}h(y, s) \in U$. Choose partitions $0 = x_0 < x_1 < \cdots < x_N = 1$ and $0 = t_0 < t_1 < \cdots < t_N = 1$ such that $|x_{i-1} - x_i| < \varepsilon$ and $|t_{i+1} - t_i| < \varepsilon$. For each t_i , we get a path $h(x, t_i)$ from I to g . Use this path to define $F(g)$. Let

$$F(g) := F_0(h(x_0, t_i)^{-1}h(x_1, t_i))F_0(h(x_1, t_i)^{-1}h(x_2, t_i)) \cdots F_0(h(x_{N-1}, t_i)^{-1}h(x_N, t_i))$$

Call $h(x_j, t_i) = h_{j,i}$ and $\tilde{h}_{r,i} = F_0(h(x_0, t_i)^{-1}h(x_1, t_i)) \cdots F_0(h(x_{r-1}, t_i)^{-1}h(x_r, t_i))$. We want to show that $h_{N,i}$ doesn't depend on i i.e. $\tilde{h}_{N,i+1} = \tilde{h}_{N,i}$. To prove this, we will prove the following identity:

$$\tilde{h}_{r,i+1} = \tilde{h}_{r,i}F_0(h_{r,i}^{-1}h_{r,i+1})$$

If we know that, take $r = N$ then

$$\tilde{h}_{N,i+1} = \tilde{h}_{N,i}F_0(h_{N,i}^{-1}h_{N,i+1}) = \tilde{h}_{N,i}F_0(I) = \tilde{h}_{N,i}$$

We want to prove by induction on r . For $r = 0$, $\tilde{h}_{0,i+1} = \tilde{h}_{0,i} = I$ so this is true for $r = 0$. Suppose this holds for r and consider $r + 1$.

$$\begin{aligned}
\tilde{h}_{r+1,i+1} &= \tilde{h}_{r,i+1}F_0(h_{r,i+1}^{-1}h_{r+1,r+1}) \\
&= \tilde{h}_{r,i}F_0(h_{r,i}^{-1}h_{r,i+1})F_0(h_{r,i+1}^{-1}h_{r+1,i+1}) \\
&= \tilde{h}_{r,i}F_0(h_{r,i}^{-1}h_{r+1,i+1}) \\
&= \tilde{h}_{r,i}F_0(h_{r,i}^{-1}h_{r+1,i})F_0(h_{r+1,i}^{-1}h_{r+1,i+1}) \\
&= \tilde{h}_{r+1,i}F_0(h_{r+1,i}^{-1}h_{r+1,i+1})
\end{aligned}$$

□

6 Representation Theory

Definition 6.1. Let G be a group. A representation of G is pair (V, R) where V is a finite dimensional vector space over $\mathbb{F}$ representation space. And $R : G \rightarrow GL(V) = \text{Aut}(V)$ is an automorphism, $GL(V)$ is the space of invertible maps $V \rightarrow V$.

Definition 6.2. A representation of G is a pair $(V, *)$ where V is a finite dimensional vector space over $\mathbb{F}$ and $*$ is an action of G on V such that $\forall g \in G, v \mapsto g * v$ is a linear map $V \rightarrow V$ i.e. for any $v, w \in V, \lambda \in \mathbb{F}$, $g * (v + \lambda w) = g * v + \lambda g * w$.

Remark. The above two definitions are equivalent. We want to define a group homomorphism $R : G \rightarrow GL(V)$ i.e. $R(g) : V \rightarrow V$ linear map and $R(g)v = g * v$.

By definition

$$\begin{aligned} R(gh)v &= (gh) * v = g * (h * v) \\ &= R(g)(R(h)v) \\ &= R(g)R(h)v \implies R(gh) = R(g)R(h) \end{aligned}$$

Since $R(g)R(g^{-1})(v) = R(gg^{-1})(v) = I * v = v$ so $R(g^{-1})$ is the inverse of $R(g)$.

Now let's start $R : G \rightarrow GL(V)$ and we want to construct an action $*$ of G on V . Let $g * v = R(g)v$, by definition

$$\begin{aligned} (gh) * v &= R(gh)v = R(g)R(h)v \\ &= g * (h * v) \end{aligned}$$

and $I * v = R(I)v = v$ so it's indeed a group action. In addition, linearity is trivial since it's a group homomorphism.

Examples.

- (1) Trivial representation: Let V be any vector space then the trivial representation of G is (V, R) where $R(g)v = v$ i.e. $R(g) = I$ for all $g \in G$.
- (2) Standard representation: Suppose $G \leq GL(n, \mathbb{F})$ then $(\mathbb{F}^n, R)$ where $R(g) = g$ is the standard representation i.e. $R(g)v = gv$ is the matrix multiplication.
- (3) Let $G \leq GL(n, \mathbb{F})$, take $V = \mathbb{F}$ and $R(g) = \det(g) \in GL(V)$ which is indeed a representation.
- (4) The adjoint representation: Let G be a matrix Lie group and $\mathfrak{g}$ is the Lie algebra. Let $Ad : G \rightarrow GL(\mathfrak{g})$ and $Ad(g)X = gXg^{-1}$.
- (5) Let $G = GL(2, \mathbb{R})$ and

$$V = \{f(x, y) = ax^3 + bx^2y + cxy^2 + dy^3 \mid a, b, c, d \in \mathbb{R}\}$$

Let $(g * f)(v) = f(g^{-1}v)$ then we claim that $(V, *)$ is a representation since

$$\begin{aligned} ((gh) * f)v &= f((gh)^{-1}v) = f(h^{-1}g^{-1}v) \\ &= h * f(g^{-1}v) = (g * (h * f))v \end{aligned}$$

and $(I * f)(v) = f(v)$ i.e. $I * f = I$.

- (6) Direct Sum: Given $(V, *_V)$ $(W, *_W)$ representation of a group G . Then we can define $V \oplus W = \left\{ \begin{pmatrix} v \\ w \end{pmatrix} \mid v \in V, w \in W \right\}$. We define

$$g *_{V \oplus W} \begin{pmatrix} v \\ w \end{pmatrix} = \begin{pmatrix} g *_V v \\ g *_W w \end{pmatrix}$$

If $R_v : G \rightarrow GL(V)$ where V is an n -dimensional vector space over $\mathbb{F}$ and $R_w : G \rightarrow GL(W)$ is an m -dimensional vector space over $\mathbb{F}$. Then

$$R_{V \oplus W}(g) = \begin{pmatrix} R_V(g) & 0 \\ 0 & R_W(g) \end{pmatrix}$$

where $R_{V \oplus W} : G \rightarrow GL(n + m, \mathbb{F})$ and this is a representation.

- (7) Let V, W be vector spaces and let

$$\text{Hom}(V, W) = \{\text{linear maps } \varphi : V \rightarrow W\}$$

then

$$(g *_{\text{Hom}(V, W)} \varphi)v = g *_W \varphi(g^{-1} *_V v)$$

Example. Let $G = GL(2, \mathbb{F})$ and $V = \mathbb{F}^2$. Now consider

$$V^* = \{ax + by \mid a, b \in \mathbb{F}\} = \{\varphi : \mathbb{F}^2 \rightarrow \mathbb{F}, \begin{pmatrix} x \\ y \end{pmatrix} \mapsto ax + by\}$$

Let $g = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} \in GL(2, \mathbb{F})$ then

$$(g *_V \varphi) \begin{pmatrix} x \\ y \end{pmatrix} = \varphi(g^{-1} *_V \begin{pmatrix} x \\ y \end{pmatrix}) = \varphi(g^{-1} \begin{pmatrix} x \\ y \end{pmatrix}) = \varphi \left(\begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \right) = \varphi \begin{pmatrix} -5x + 3y \\ 2x - y \end{pmatrix}$$

Moreover, take $\{b_1^*, b_2^*\}$ be a basis for V^* where

$$b_1^* \begin{pmatrix} x \\ y \end{pmatrix} = x, \text{ and } b_2^* \begin{pmatrix} x \\ y \end{pmatrix} = y$$

Then

$$(g *_V b_1^*) \begin{pmatrix} x \\ y \end{pmatrix} = b_1^* \begin{pmatrix} -5x + 3y \\ 2x - y \end{pmatrix} = -5x + 3y$$

$$(g *_V b_2^*) \begin{pmatrix} x \\ y \end{pmatrix} = b_2^* \begin{pmatrix} -5x + 3y \\ 2x - y \end{pmatrix} = 2x - y$$

Note that

$$(R_{V^*}(g))_{\{b_1^*, b_2^*\}} = \begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix} = (g^T)^{-1}$$

Example. Tensor product: Let V, W be representations of a group G . We will define $V \otimes W$ to be the tensor product of V, W . Let $\{b_1, \dots, b_n\}$ be a basis V and $\{c_1, \dots, c_m\}$ be a basis W . Then $V \otimes W$ will have the symbols $b_i \otimes c_j$ as basis vectors i.e. each elements $V \otimes W$ can be written as

$$\sum_{i,j} t_{ij} b_i \otimes c_j, \quad t_{ij} \in \mathbb{F}$$

Moreover, $\otimes$ is bilinear i.e.

$$(v_1 + \alpha v_2) \otimes w = v_1 \otimes w + \alpha v_2 \otimes w, \text{ and } v \otimes (w_1 + \beta w_2) = v \otimes w_1 + \beta v \otimes w_2$$

Now we define

$$g *_V \otimes W (x \otimes y) = (g *_V x, g *_W y)$$

which is a representation.

Remark. Not all elements $V \otimes W$ can be written as $x \otimes y$. E.g. $e_1 \otimes e_2 + e_2 \otimes e_1 \in V \otimes W$. Then elements $x \otimes y$ are called pure tensors.

Definition 6.3. Consider the action S_n on $V^{\otimes n}$ which is defined on pure tensors by

$$\sigma(v_1 \otimes \dots \otimes v_n) = v_{\sigma(1)} \otimes \dots \otimes v_{\sigma(n)}, \sigma \in S_n$$

The n^{th} symmetric power of V is the subspace $Sym^n(V) \subset V^{\otimes n}$ consisting of tensors which are fixed by every $\sigma \in S_n$ under this action.

Example. Suppose that V is a vector space with basis $\{b_1, b_2\}$. Then tensor $b_1 \otimes b_2 + b_2 \otimes b_1$ is an element of Sym^2 but the tensor $b_1 \otimes b_2$ is not.

Definition 6.4. We define the averaging map to be

$$Av : V^{\otimes n} \rightarrow V^{\otimes n}, Av(v) = \frac{1}{n!} \sum_{\sigma \in S_n} \sigma(v)$$

Note that Av is a projection onto its image $Sym^n(V) \subset V^{\otimes n}$.

Example. If a vector space with basis $\{b_1, b_2\}$, then $Av(b_1 \otimes b_2) = \frac{1}{2}(b_1 \otimes b_2 + b_2 \otimes b_1)$.

Proposition 6.5. If $(V, *_V)$ is a representation of G , then $Sym^n(V)$ is a subrepresentation of $V^{\otimes n}$.

Definition 6.6. Consider the action of S_n on $V^{\otimes n}$ which is defined on the pure tensors by

$$\tau(v_1 \otimes \dots \otimes v_n) = (-1)^{|\tau|} v_{\tau(1)} \otimes \dots \otimes v_{\tau(n)}, \quad \tau \in S_n$$

where $|\tau|$ is the sign of the permutation. The n^{th} exterior power of V is the subspace $\Lambda^n(V) \subset V^{\otimes n}$ consisting of tensors which are fixed by every $\tau \in S_n$ under this action.

Example. Suppose V is a vector space with basis $\{b_1, b_2\}$. The tensor $b_1 \otimes b_2 - b_2 \otimes b_1$ is an element of Λ^2 but the tensor $b_1 \otimes b_2$ is not.

Definition 6.7. We define the alternating map

$$Alt : V^{\otimes n} \rightarrow V^{\otimes n}, Alt(v) = \frac{1}{n!} \sum_{\tau \in S_n} \tau(v)$$

Note that Alt is a projection onto its image $\Lambda^n(V) \subset V^{\otimes n}$.

Proposition 6.8. If $(V, *)$ is a representation of G , then $\Lambda^n(V)$ is a subrepresentation of $V^{\otimes n}$.

Definition 6.9. We introduce some new notation for exterior powers which will be convenient

$$v_1 \wedge \cdots \wedge v_n = Alt(v_1 \otimes \cdots \otimes v_n).$$

For example $x \wedge y = \frac{1}{2}(x \otimes y - y \otimes x)$. Moreover one can show that Λ has the following properties:

$$v \wedge w = -w \wedge v, v \wedge v = 0$$

Example. Let $G = GL(2, \mathbb{C})$ and $(V, *_V) = (\mathbb{C}^2, St)$ be the standard representation. Take $b_1 = (1, 0)^T$ and $b_2 = (0, 1)^T$ be a basis for V . Let $g = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in G$. We will compute $g * (b_1 \otimes b_2)$, $g * (b_1 b_2)$ and $g * (b_1 \wedge b_2)$. First note that

$$\begin{aligned} g * b_1 &= g b_1 = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} = b_1 + 3b_2 \\ g * b_2 &= g b_2 = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 2b_2 + 4b_1 \end{aligned}$$

Then

$$\begin{aligned} g * (b_1 \otimes b_2) &= (g * b_1) \otimes (g * b_2) \\ &= (b_1 + 3b_2) \otimes (2b_1 + 4b_2) \\ &= 2b_1 \otimes b_1 + 4b_1 \otimes b_2 + 6b_2 \otimes b_1 + 12b_2 \otimes b_2 \\ g * (b_1 b_2) &= (g * b_1)(g * b_2) \\ &= (b_1 + 3b_2)(2b_1 + 4b_2) \\ &= 2b_1^2 + 10b_1 b_2 + 12b_2^2 \\ g * (b_1 \wedge b_2) &= (g * b_1) \wedge (g * b_2) \\ &= (b_1 + 3b_2) \wedge (2b_1 + 4b_2) \\ &= 2b_1 \wedge b_1 + 4b_1 \wedge b_2 + 6b_2 \wedge b_1 + 12b_2 \wedge b_2 \\ &= 4b_1 \wedge b_2 - 6b_1 \wedge b_2 = -2b_1 \wedge b_2 \end{aligned}$$

Definition 6.10. A linear map $\Phi : V \rightarrow W$ is a morphism of representations if for all $v \in V$, $g \in G$ we have

$$\Phi(g *_V v) = g *_W (\Phi(v))$$

where $(V, *_V)$ and $(W, *_W)$ are representations of some group G i.e. the following diagram commutes

$$\begin{array}{ccc} V & \xrightarrow{\Phi} & W \\ \downarrow R_V(g) & & \downarrow R_W(g) \\ V & \xrightarrow{\Phi} & W \end{array}$$

Moreover, if Φ is also invertible we call it an isomorphism of representations.

Given a matrix Lie group G , what are the isomorphism classes of smooth representations of G . Here smooth refers to the homomorphism $R : G \rightarrow GL(V)$ to be smooth.

Definition 6.11. A subspace $W \leq V$ is called a subrepresentation of V if for all $g \in G$, $w \in W$ we have $g * w \in W$ or equivalently $(R(g)w \in W)$.

Remark. A homomorphism $R|_W : G \rightarrow GL(W)$ where $R|_W R(g) = R(g)|_W$ then $(W, R(g)|_W)$ is a representation of G .

Example. Give a representation V of G , we always have $0, V$ subrepresentations.

Definition 6.12. A subrepresentation W , $W \neq 0$ or V is called a proper subrepresentation.

Example. Consider $G = GL(2, \mathbb{R})$, $V = \mathfrak{gl}(2, \mathbb{R})$ with $Ad(g)X = gXg^{-1}$ for all $g \in G$ and $X \in V$. Let

$$W = \left\{ \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} : x \in \mathbb{R} \right\}, \quad U = \{X \in \mathfrak{gl}(2, \mathbb{R}) : \text{Tr}(X) = 0\} = \mathfrak{sl}(2, \mathbb{R})$$

Note that

$$g * \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} = g \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} g^{-1} = \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} \in W$$

i.e. W is a subrepresentation of V . Let $X \in U$, consider

$$\text{Tr}(g * X) = \text{Tr}(gXg^{-1}) = \text{Tr}(X) = 0$$

Thus, U is also a subrepresentation of V . Note that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \frac{a+d}{2} & 0 \\ 0 & \frac{a+d}{2} \end{pmatrix} + \begin{pmatrix} \frac{a-d}{2} & b \\ c & \frac{d-a}{2} \end{pmatrix} \\ \implies V = W \oplus U$$

Definition 6.13. A representation V is called irreducible/simple if it has no proper subrepresentations.

Definition 6.14. A representation (V, R) is called semi-simple/completely decomposable if

$$V = \bigoplus_{i=1}^r V_i$$

where V_i is irreducible/simple.

Example. Let $G = SL(2, \mathbb{C})$, $V = \mathbb{C}^2$ and the standard representation $St(g)v = gv$. We claim that this is irreducible.

Proof. Let W be a non-zero subrepresentation of V . It suffices to show $W = V$. Let $v_1 \in W$ with $v_1 \neq 0$. Choose $v_2 \in V$ such that $\{v_1, v_2\}$ is a basis for V . Define $gv_1 = v_2$ and $gv_2 = -v_1$ then

$$[g_1]_{\{v_1, v_2\}} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

In particular, $\deg g = 1$ so $g \in SL(2, \mathbb{C})$. Since W is a subrepresentation of V , we must have $g * v_1 = gv_1 \in W$ since $v_1 \in W$. But $gv_1 = v_2 \in W \implies V = W$. $\square$

Example. Any 1-dimensional representation is irreducible.

Example. Tensor powers are not irreducible as they always have proper subrepresentations i.e. symmetric powers.

Remark. Suppose V is semi-simple, then we can write

$$V = V_1 \oplus \cdots \oplus V_r$$

where V_i is irreducible. Also suppose linear map $R(g)$, write is a matrix with respect to basis vectors of $V_1, \dots, V_r$, so that the basis V_i , before basis of V_j for $i < j$ then

$$[R(g)] = \begin{pmatrix} R(g)|_{v_1} & & & \\ & R(g)|_{v_2} & & \\ & & \ddots & \\ & & & R(g)|_{v_r} \end{pmatrix}$$

Then $R = R|_{v_1} \oplus \cdots \oplus R|_{v_r}$. When V_i are small the matrix is very closed to diagonal. Unfortunately not all subrepresentations are completely diagonalisable.

Example. Consider the group $(\mathbb{R}, +)$, we have a representation $\mathbb{R}^2$

$$R(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$$

This is not irreducible, we have 1-dimensional subrepresentation

$$Y = \left\{ \begin{pmatrix} y \\ 0 \end{pmatrix} : y \in \mathbb{R} \right\}$$

Suppose we can find a complementary subspace Z spanned by $\begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$ such that $z_2 \neq 0$. Then

$$\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} z_1 + xz_2 \\ z_2 \end{pmatrix} \in Z$$

i.e. there exists some $f(x)$ such that

$$f(x) \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} z_1 + xz_2 \\ z_2 \end{pmatrix}$$

But $f(x) = 1$ and $z_1 + xz_2 = z_1 \implies x = 0$ so we reach a contradiction here.

Theorem 6.15 (Weyl). Every smooth representation of a compact matrix Lie group is semi-simple.

Sketch. Suppose (V, R) is a representation of G which is compact i.e. we have a G -invariant inner-product. Then we have two options for V :

- (i) V is simple.
- (ii) V is not simple i.e. it has a proper subrepresentation say W . By the Lemma below, we have $W^\perp$ which is also

a subrepresentation and in particular $V = W \oplus W^\perp$.

Then we repeat the above process and we are done. $\square$

Theorem 6.16 (Weyl's Unitary Tricks). If G is a compact matrix Lie group then there exists a G -invariant Hermitian inner product.

Definition 6.17. Let V be vector space over $\mathbb{C}$. A Hermitian inner product on V is a function $V \times V \rightarrow \mathbb{C}$ such that

- (i) $\langle v, v \rangle$ is real, positive for all $v \in V$.
- (ii) $\langle u, v \rangle = \overline{\langle v, u \rangle}$ for all $u, v \in V$.
- (iii) $\langle u + \lambda v, w \rangle = \langle u, w \rangle + \lambda \langle v, w \rangle$ for all $u, v, w \in V$ and $\lambda \in \mathbb{C}$.

Example. Let $V = \mathbb{C}^n$ then the standard Hermitian inner product is given by

$$\left\langle \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}, \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} \right\rangle = v_1 \overline{w_1} + \cdots + v_n \overline{w_n}$$

Thus is a replacement of the dot product.

Definition 6.18. Suppose (V, R) is a representation of G . A Hermitian inner product is G -invariant if

$$\langle R(g)v, R(g)w \rangle = \langle v, w \rangle$$

for all $g \in G, v, w \in V$.

Lemma 6.19. Let (V, R) be a representation of G , with a G -invariant Hermitian inner product. If W is a subrepresentation of V then

$$W^\perp = \{v \in V : \langle v, w \rangle = 0, \quad w \in W\}$$

is a subrepresentation of V .

Proof. Let $g \in G, v \in W^\perp$ and we want to show $R(g)v \in W^\perp$. Consider $w \in W$

$$\begin{aligned} \langle R(g)v, w \rangle &= \langle R(g^{-1})R(g)v, R(g^{-1})w \rangle \\ &= \langle v, R(g^{-1})w \rangle \\ &= \langle v, w' \rangle = 0 \end{aligned}$$

Since W is a representation so $R(g^{-1})w = w'$ for some $w' \in W$. Thus, $W^\perp$ is a subrepresentation. $\square$

Example. $U(n), SU(n), O(n)$ and $U(n)$ are compact and we are going to study the representations of these groups

7 Lie algebra representations

Given a smooth representation $R : G \rightarrow GL(n, \mathbb{C})$ we get a Lie algebra homomorphism $R_* : \mathfrak{g} \rightarrow \mathfrak{gl}(n, \mathbb{C})$, which has a linear map such that $R_*[X, Y] = [R_*X, R_*Y]$ for all $X, Y \in \mathfrak{g}$.

Definition 7.1. Let $\mathfrak{g}$ be a Lie algebra over $\mathbb{C}$. A Lie algebra representation is a pair (V, r) where V is a finite dimensional vector space and $r : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ is a Lie algebra homomorphism. $\mathfrak{g}(V)$ consists of all linear maps $V \rightarrow V$.

Remark. A key property $R(\exp X) = \exp(R_*X)$ where R is a smooth Lie group homomorphism and $X \in \mathfrak{g}$. Then we differentiate to get

$$R_*(X) = \frac{d}{dt}\bigg|_{t=0} R(\exp tX)$$

If G is simply connected, then R_* give rise to Lie group homomorphism.

Examples.

- (1) Consider the standard representation. Let $(V, R) = (\mathbb{C}^n, St)$ standard representation of Lie group G where $St(g)v = gv$ for all $v \in V$. Let $X \in \mathfrak{g}$, then

$$r(X)v = \frac{d}{dt}\bigg|_{t=0} St(\exp tX)v = Xv$$

i.e. the standard representation has Lie algebra representation which is also matrix multiplication.

- (2) Adjoint representation. Let $(V, R) = (\mathfrak{g}, Ad)$ be the adjoint representation of a Lie group of G . Recall that G acts in the following way $Ad(g)Y = gYg^{-1}$. Let $X, Y \in \mathfrak{g}$. Then

$$\begin{aligned} r(X)Y &= \frac{d}{dt}\bigg|_{t=0} Ad(\exp(tX)Y) = \frac{d}{dt}\bigg|_{t=0} ((\exp tX)Y \exp(-tX)) \\ &= \frac{d}{dt}\bigg|_{t=0} ((I + tX)Y(I - tX) + o(t)) = \frac{d}{dt}\bigg|_{t=0} (Y + t(XY - YX) + o(t)) \\ &= XY - YX = [X, Y] = ad(X)Y \end{aligned}$$

- (3) Tensor product. Let (V, R_V) and (W, R_W) be representations of G . Then consider the tensor product representation $(V \otimes W, R)$. For $X \in \mathfrak{g}$ and $v \otimes w \in V \otimes W$:

$$\begin{aligned} r(X)v_1 \otimes v_2 &= \frac{d}{dt}\bigg|_{t=0} R(\exp(tX))(v \otimes w) = \frac{d}{dt}\bigg|_{t=0} (R_V(\exp(tX))v) \otimes (R_W(\exp(tX))w) \\ &= \frac{d}{dt}\bigg|_{t=0} (\exp(tr_V(X))v) \otimes (\exp(tr_W(X))w) = \frac{d}{dt}\bigg|_{t=0} ((I + tr_V(X))v) \otimes ((I + tr_W(X))w) \\ &= \frac{d}{dt}\bigg|_{t=0} (v \otimes w + t(r_V(X)v \otimes w + v \otimes r_W(X)w) + o(t)) = r_V(X)v \otimes w + v \otimes r_W(X)w \end{aligned}$$

Definition 7.2. If V is a finite dimensional real vector space, then the complexification of V , denoted by $V^{\mathbb{C}}$, is the vector space of formal linear combinations $v_1 + iv_2$, with $v_1, v_2 \in V$. This becomes a complex vector space if we define

$$i(v_1 + iv_2) = -v_2 + iv_1$$

Suppose $\mathfrak{g}$ is a finite dimensional Lie algebra over $\mathbb{R}$, and let $\mathfrak{g}^{\mathbb{C}}$ be its complexification, then the Lie bracket extends in a complex linear way:

$$[v_1 + iw_1, v_2 + iw_2] = [v_1, v_2] - [w_1, w_2] + i([v_1, w_2] + [w_1, v_2])$$

Examples.

- (1) $\mathfrak{gl}(n, \mathbb{R})^{\mathbb{C}} = \mathfrak{gl}(n, \mathbb{C})$.
- (2) $\mathfrak{sl}(n, \mathbb{R})^{\mathbb{C}} = \mathfrak{sl}(n, \mathbb{C})$.
- (3) We have $u(n)^{\mathbb{C}} = \mathfrak{gl}(n, \mathbb{C})$.

Lemma 7.3. There is one-to-one correspondence between complex representations of $\mathfrak{g}$ and complex-linear complex representations of $\mathfrak{g}^{\mathbb{C}}$.

Proof. If $r : \mathfrak{g} \rightarrow \mathfrak{gl}(n, \mathbb{C})$ is a real-linear Lie algebra homomorphism $r^{\mathbb{C}} : \mathfrak{g}^{\mathbb{C}} \rightarrow \mathfrak{gl}(n, \mathbb{C})$ by $r^{\mathbb{C}}(X + iY) = r(X) + ir(Y)$. Conversely, if $s : \mathfrak{g}^{\mathbb{C}} \rightarrow \mathfrak{gl}(n, \mathbb{C})$ is a complex linear map then $s(X + iY) = s(X) + is(Y)$, so setting $r = s|_{\mathfrak{g}}$, we get $r^{\mathbb{C}} = s$. $\square$

Corollary 7.4. V is a complex representation of $\mathfrak{g}$ and hence also a complex linear representation of $\mathfrak{g}^{\mathbb{C}}$. Then

- (1) V is simple as representation of $\mathfrak{g}$ if and only if V is simple as a representation of $\mathfrak{g}^{\mathbb{C}}$.
- (2) V is semi-simple as a representation of $\mathfrak{g}$ if and only if V is semi-simple as a representation of $\mathfrak{g}^{\mathbb{C}}$.

8 Representations of $U(1)$ and $SU(2)$

Lemma 8.1 (Schur's). Let (V, R) be a simple representation of a group G over $\mathbb{C}$. If $L : V \rightarrow V$ is a linear map which commutes with $R(g)$ for all $g \in G$, then $L = \lambda I$ for some $\lambda \in \mathbb{C}$.

Proof. Since $\mathbb{C}$ is algebraically closed, the characteristic polynomial of L has at least one eigenvalue $\lambda \in \mathbb{C}$ for which the eigenspace

$$V_\lambda = \{v \in V : Lv = \lambda v\}$$

is not 0. Since V is simple. Then V_λ is a subrepresentation of V , since it is closed under the action of the representation. Let $v \in V_\lambda$, then

$$\begin{aligned} L(R(g)v) &= R(g)L(v) = R(g)\lambda v \\ &= \lambda R(g)v \end{aligned}$$

Therefore, $R(g)v \in V_\lambda$. But since V is simple and $V_\lambda \neq 0$ we must have $V = V_\lambda$. Since everything is an eigenvector with an eigenvalue λ , we must have $Lv = \lambda Iv$. $\square$

Corollary 8.2. If G is an abelian group then every simple representation of G over $\mathbb{C}$ is 1-dimensional.

Proof. Let (V, R) be a simple representation of G an abelian group. Let $g \in G$, then $R(g)$ commutes with $R(h)$ for all $h \in G$. Since R is a homomorphism and G is abelian:

$$R(g)R(h) = R(gh) = R(hg) = R(h)R(g).$$

By Schur's lemma $R(g) = \lambda I$, for some $\lambda \in \mathbb{C}^\times$. Choose $v \in V$, $v \neq 0$ then $R(g)v = \lambda v$ and $\mathbb{C}v$ is a subrepresentation and since V is simple and $\mathbb{C}v \neq 0$, we must have $V = \mathbb{C}v$, that is 1-dimensional. $\square$

Lemma 8.3. Any simple representation of $U(1)$ is 1-dimensional and given by $R(e^{i\theta}) = e^{im\theta}$ for some $m \in \mathbb{Z}$.

Proof. Let (V, R) be a simple representation of $U(1)$. First note that $U(1)$ is abelian, hence by previous corollary every simple representation of $U(1)$ is 1-dimensional. It remains to show that any simple representation of $U(1)$ is given by $R(e^{i\theta}) = e^{im\theta}$ for some $m \in \mathbb{Z}$.

We also have $R : U(1) \rightarrow GL(1, \mathbb{C}) = \mathbb{C}^\times$ to be a homomorphism. Let $e^{i\theta} \in U(1)$, then by Schur's lemma $R(e^{i\theta}) = \lambda I$ for some $\lambda \in \mathbb{C}^\times$. Notice that $\lambda \neq 0$ since $R(e^{i\theta}) \in GL(V)$ so it is an invertible map. Moreover, $U(1)$ is compact, so there is an invariant Hermitian inner product. Namely,

$$\begin{aligned} \langle v, v \rangle &= \langle R(e^{i\theta}v), R(e^{i\theta}v) \rangle \\ &= \langle \lambda v, \lambda v \rangle = |\lambda|^2 \langle v, v \rangle \end{aligned}$$

Hence $|\lambda|^2 = 1$ then $\lambda \in U(1)$. That is R is a homomorphism $U(1) \rightarrow U(1)$ and we know that these are precisely $e^{im\theta}$ for $m \in \mathbb{Z}$. $\square$

Remark. So every representation $(\mathbb{C}^n, R)$ of $U(1)$ is semi-simple $\mathbb{C}^n = V_1 \oplus \dots \oplus V_n$, where each V_i is 1-dimensional subrepresentation with

$$[R(e^{i\theta})] = \begin{pmatrix} e^{im_1\theta} & \dots & 0 \\ & \ddots & \\ 0 & \dots & e^{im_n\theta} \end{pmatrix}$$

That is $R = R_1 \oplus \dots \oplus R_n$ such that $R_k(e^{i\theta}) = e^{im_k\theta}$. We will call the integers $m_1, \dots, m_n$ the weights of V .

Definition 8.4. The weight space of weight k in V is

$$V[k] = \{v \in V : R(e^{i\theta})v = e^{ik\theta}v\}$$

vectors of weight k .

Example. Suppose $V = V[2] \oplus V[3] \oplus V[3] \oplus V[-1] \oplus V[2]$. The weights of V are $-1, 2, 2, 3, 3$.

Example. Let (V, R) be a complex representation of $U(1)$, with $R : U(1) \rightarrow GL(2, \mathbb{C})$ given by $R(e^{i\theta}) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \in GL(2, \mathbb{C})$. The characteristic polynomial of this matrix is

$$\det \begin{pmatrix} \cos \theta - \lambda & -\sin \theta \\ \sin \theta & \cos \theta - \lambda \end{pmatrix} = \lambda^2 - 2\lambda \cos \theta + 1$$

so eigenvalues are $\lambda = e^{\pm i\theta}$. Therefore the weights of this representation are ± 1 . The eigenvectors are $(i, 1)$ and $(-i, 1)$. There are therefore our vectors $v_1 \in V_1 = V[1]$ and $v_2 \in V_2 = V[-1]$. With respect to this basis of eigenvectors

$$R(e^{i\theta}) = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}$$

That is $V = V[1] \oplus V[-1]$.

Note that $SU(2)$ not abelian, so simple representations are not just 1-dimensional complex vector space. However we can define

$$T = \left\{ \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} \mid \theta \in \mathbb{R} \right\}$$

Given (R, V) of $SU(2)$, we consider $R|_T : U(1) \rightarrow GL(V)$ homomorphism which gives us a representation of $U(1)$

$$V = V_1 \oplus \cdots \oplus V_n$$

and

$$V[m] = \{v \in V \mid R \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} v = e^{im\theta} v\}$$

and at the level of Lie algebra

$$V[m] = \{v \in V \mid r(iH)v = imv\}$$

at the level of the complexification of the Lie algebra

$$V[m] = \{v \in V \mid r^{\mathbb{C}}(H)v = mv\}$$

where $H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Example. Let $(V, R) = (su(2), Ad)$ of $SU(2)$ then

$$R : SU(2) \rightarrow GL(su(2))$$

$$R(g)M_v = gM_v g^{-1}$$

where $M_v = \begin{pmatrix} ix & y + iz \\ -y + iz & -ix \end{pmatrix} \in su(2)$. And

$$r : su(2) \rightarrow \mathfrak{gl}(3, \mathbb{R})$$

$$r \begin{pmatrix} ix & y + iz \\ -y + iz & -ix \end{pmatrix} = \begin{pmatrix} 0 & -2x & 2y \\ 2z & 0 & -2x \\ 2y & 2x & 0 \end{pmatrix}$$

Consider $R|_T : U(1) \rightarrow GL(su(2)^{\mathbb{C}})$ then

$$\begin{aligned} R(\exp \begin{pmatrix} ix & 0 \\ 0 & -ix \end{pmatrix}) &= \exp \left(r \begin{pmatrix} ix & 0 \\ 0 & -ix \end{pmatrix} \right) \\ &= \exp \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -2x \\ 0 & 2x & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 2x & -\sin 2x \\ 0 & \sin 2x & \cos 2x \end{pmatrix} \end{aligned}$$

This matrix diagonalises to

$$\begin{pmatrix} 1 & & \\ & e^{2ix} & \\ & & e^{-2ix} \end{pmatrix}$$

i.e. $R|_T = R_0|_T \oplus R_2|_T \oplus R_{-2}|_T$ and

$$R_0|_T(g)v = v$$

$$R_2|_T(g)v = e^{2ix}v$$

$$R_{-2}|_T(g)v = e^{-2ix}v$$

Therefore, the weights are $0, 2, -2$. Then $V|_T = V[0] \oplus V[2] \oplus V[-2]$.

Example. Let (V, S) be representation given by second symmetric power of the symmetric representation. $S = Sym^2(\mathbb{C}^2)$. Let e_1, e_2 be the standard basis for $\mathbb{C}^2$ then $Sym^2(\mathbb{C}^2)$ has basis e_1^2, e_1e_2, e_2^2 . Consider $Sym^2(\mathbb{C}^2)|_T : U(1) \rightarrow GL((\mathbb{C}^2)^{\otimes 2})$ satisfies

$$Sym^2 \left(\begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} \right) e_1^2 = e^{i2\theta} e_1^2$$

$$Sym^2 \left(\begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} \right) e_1e_2 = e_1e_2$$

$$Sym^2 \left(\begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} \right) e_2^2 = e^{-2i\theta} e_2^2$$

By writing down the diagonal matrix the weights are $0, 2, -2$ same as previously.

Lemma 8.5. Let H, X, Y be the basis elements of $\mathfrak{sl}(2, \mathbb{C})$ where

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

then

- (1) $r^{\mathbb{C}}(H)v = mv, v \in V[m]$.
- (2) $r^{\mathbb{C}}(X)v \in V[m+2], v \in V[m]$.
- (3) $r^{\mathbb{C}}(Y)v \in V[m-2], v \in V[m]$.

Proof.

- (1) By definition $r^{\mathbb{C}}(H) = r^{\mathbb{C}} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and the result

$$r^{\mathbb{C}} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} v = mv$$

follows by definition.

- (2) $v \in V[m]$ if and only if $r^{\mathbb{C}}(H)v = mv$. Consider $r^{\mathbb{C}}(H)(r^{\mathbb{C}}(X)v)$, we want to show that this equals to $(m+2)r^{\mathbb{C}}(X)v$. We know that

$$r^{\mathbb{C}}([H, X]) = [r^{\mathbb{C}}(H), r^{\mathbb{C}}(X)] = r^{\mathbb{C}}(H)r^{\mathbb{C}}(X) - r^{\mathbb{C}}(X)r^{\mathbb{C}}(H)$$

Thus,

$$\begin{aligned} r^{\mathbb{C}}(2X) &= r^{\mathbb{C}}(H)r^{\mathbb{C}}(X) - r^{\mathbb{C}}(X)r^{\mathbb{C}}(H) \\ 2r^{\mathbb{C}}(X) + r^{\mathbb{C}}(X)r^{\mathbb{C}}(H) &= r^{\mathbb{C}}(H)r^{\mathbb{C}}(X) \\ \implies 2r^{\mathbb{C}}(X)v + r^{\mathbb{C}}(X)r^{\mathbb{C}}(H)v &= r^{\mathbb{C}}(H)r^{\mathbb{C}}(X)v \\ \implies (2+m)r^{\mathbb{C}}(X)v &= r^{\mathbb{C}}(H)r^{\mathbb{C}}(X)v \end{aligned}$$

Therefore, $r^{\mathbb{C}}(X)v \in V[m+2]$.

- (3) To show $r^{\mathbb{C}}(Y)v \in V[m-2]$, we consider

$$r^{\mathbb{C}}(H)(r^{\mathbb{C}}(Y)v) = (m-2)r^{\mathbb{C}}(Y)v$$

using the fact that $[H, Y] = -2Y$.

□

Suppose we are given a weight diagram, we say m is the highest weight if m is the largest integer such that $V[m] \neq \{0\}$. Pick $v \in V[m]$, for m highest weight then

$$r^{\mathbb{C}}(H)v = mv$$

$$r^{\mathbb{C}}(X)v \in V[m+2] \quad r^{\mathbb{C}}(X)v = 0$$

Moreover, if we are considering a subrepresentation of V then this needs to be closed under the action of $r^{\mathbb{C}}(H), r^{\mathbb{C}}(X), r^{\mathbb{C}}(Y)$. We consider the following sequence

$$v, r^{\mathbb{C}}(Y)v, r^{\mathbb{C}}(Y)^2v, \dots, r^{\mathbb{C}}(Y)^nv$$

of elements of decreasing weight since each $r^{\mathbb{C}}(Y)$ reduces the weight by 2. Therefore, $r^{\mathbb{C}}(Y)^kv \in V[m-2k]$. Let n be the largest integer such that $r^{\mathbb{C}}(Y)^nv \neq 0$ but $r^{\mathbb{C}}(Y)^{n+1}v = 0$.

Lemma 8.6. Let U be spanned by $\{v, r^{\mathbb{C}}(Y)v, \dots, r^{\mathbb{C}}(Y)^nv\}$. Then $U \subset V$ is a subrepresentation of V .

Proof. To check that U is a subrepresentation we need to show U is closed under the action of $r^{\mathbb{C}}(H), r^{\mathbb{C}}(X)$ and $r^{\mathbb{C}}(Y)$.

Let $U = \text{Span}\{v, r^{\mathbb{C}}(Y)v, \dots, r^{\mathbb{C}}(Y)^nv\}$ we want to show that

$$r^{\mathbb{C}}(Y)(r^{\mathbb{C}}(Y)^kv) = r^{\mathbb{C}}(Y)^{k+1}v$$

$$r^{\mathbb{C}}(X)(r^{\mathbb{C}}(Y)^kv) = (m+1-k)kr^{\mathbb{C}}(Y)^{k-1}v$$

$$r^{\mathbb{C}}(H)(r^{\mathbb{C}}(Y)^kv) = (m-2k)r^{\mathbb{C}}(Y)^kv$$

The first equation is trivial. For the third equation, $r^{\mathbb{C}}(H)r^{\mathbb{C}}(Y)^nv = (m-2k)r^{\mathbb{C}}(Y)^nv$ since $r^{\mathbb{C}}(Y)^kv \in V[m-2k]$. It is enough to show the second equation and we will prove by induction. For $k=0$, $r^{\mathbb{C}}(X)v = 0$ so the statement is true for $k=0$. Suppose it's true for k i.e.

$$r^{\mathbb{C}}(X)(r^{\mathbb{C}}(Y)^kv) = (m+1-k)kr^{\mathbb{C}}(Y)^{k-1}v$$

Consider $r^{\mathbb{C}}(X)(r^{\mathbb{C}}(Y)^{k+1}v) = (r^{\mathbb{C}}(X)r^{\mathbb{C}}(Y))(r^{\mathbb{C}}(Y)^kv)$. Notice $[r^{\mathbb{C}}(X), r^{\mathbb{C}}(Y)] = r^{\mathbb{C}}([X, Y]) = r^{\mathbb{C}}(H)$. Therefore,

$$\begin{aligned} r^{\mathbb{C}}(X)r^{\mathbb{C}}(Y)^{k+1}v &= (r^{\mathbb{C}}(H) + r^{\mathbb{C}}(Y)r^{\mathbb{C}}(X))r^{\mathbb{C}}(Y)^kv \\ &= r^{\mathbb{C}}(H)r^{\mathbb{C}}(Y)^kv + r^{\mathbb{C}}(Y)r^{\mathbb{C}}(X)r^{\mathbb{C}}(Y)^kv \\ &= (m-2k)r^{\mathbb{C}}(Y)^kv + r^{\mathbb{C}}(Y)(m+1-k)kr^{\mathbb{C}}(Y)^{k-1}v \\ &= (m+1-(k+1))(k+1)r^{\mathbb{C}}(Y)^kv \end{aligned}$$

Therefore, U is a subrepresentation of V .

□

The simple representations with highest weight m are completely determined by m (up to isomorphism).

Lemma 8.7. The highest weight m is equal to the largest integer n such that $r^{\mathbb{C}}(Y)^n \neq 0$ but $r^{\mathbb{C}}(Y)^{n+1} = 0$.

Proof. To see that $m = n$ we look at $r^{\mathbb{C}}(X)r^{\mathbb{C}}(Y)^{n+1}v = (m-k)(k+1)r^{\mathbb{C}}(Y)^nv$ and we know that $r^{\mathbb{C}}(Y)^nv \neq 0$ and $n+1 \neq 0$ thus $m-n=0$ i.e. $m=n$. $\square$

Theorem 8.8 (Classification theorem of simple representations of $SU(2)$). Any simple representation of $SU(2)$ is isomorphic to $Sym^m(\mathbb{C}^2)$ for some $m \in \mathbb{N}_0$ and has weight diagram $-m, -m+2, \dots, m$.

Theorem 8.9 (Complete reducibility of $SU(2)$ representation). Any finite dimensional $SU(2)$ representation is a direct sum of the simple representation described above.

Examples.

- (1) Trivial representation

$$SU(2) \rightarrow GL(1, \mathbb{C}) = \mathbb{C}^\times$$

$$\begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix} \mapsto 1$$

This is a 1-dimensional $\mathbb{C}$ -representation with weight 0.

- (2) Let

$$St : SU(2) \rightarrow GL(2, \mathbb{C})$$

$$\begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix} \mapsto \text{multiply by } \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}$$

The basis of $\mathbb{C}^2$ is e_1, e_2 and

$$r^{\mathbb{C}}(H)e_1 = He_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \cdot e_1$$

so $e_1 \in V[1]$ and similarly $e_2 \in V[-1]$.

- (3) $\mathbb{C}^2 \otimes \mathbb{C}^2$ which has the basis $e_1 \otimes e_1, e_1 \otimes e_2, e_2 \otimes e_1, e_2 \otimes e_2$. Note that

$$r^{\mathbb{C}}(H)(e_1 \otimes e_1) = r^{\mathbb{C}}(H)e_1 \otimes e_1 + e_1 \otimes r^{\mathbb{C}}(H)e_1 = 2e_1 \otimes e_1$$

Similarly, we have

$$r^{\mathbb{C}}(H)(e_2 \otimes e_2) = r^{\mathbb{C}}(H)e_2 \otimes e_2 + e_2 \otimes r^{\mathbb{C}}(H)e_2 = -2e_2 \otimes e_2$$

and other two terms are zero. Note that $\mathbb{C}^2 \otimes \mathbb{C}^2$ is not simple but we see that $\mathbb{C}^2 \otimes \mathbb{C}^2 = Sym^2(\mathbb{C}^2) \oplus \mathbb{C}$. In addition, $r^{\mathbb{C}}(Y)(e_1 \otimes e_1) = r^{\mathbb{C}}(Y)e_1 \otimes e_1 + e_1 \otimes r^{\mathbb{C}}(Y)e_1 = e_2 \otimes e_1 + e_1 \otimes e_2$.

- (4) $Sym^2(Sym^2(\mathbb{C}^2))$, $Sym^2(\mathbb{C}^2)$ has basis α, β, γ then $Sym^2(Sym^2(\mathbb{C}^2))$ has basis $\alpha^2, \beta^2, \gamma^2, \alpha\gamma, \alpha\beta, \beta\gamma$. Note that

$$r^{\mathbb{C}}(H)\alpha^2 = (r^{\mathbb{C}}(H)\alpha)\alpha + \alpha(r^{\mathbb{C}}(H)\alpha) = 2\alpha^2 + 2\alpha^2 = 4\alpha^2 \in V[4]$$

Similarly,

$$r^{\mathbb{C}}(H)\alpha\beta = 2\alpha\beta \in V[2]$$

$$r^{\mathbb{C}}(H)\alpha\gamma = 0 \in V[0]$$

etc. Then $Sym^2(Sym^2(\mathbb{C}^2)) = Sym^4(\mathbb{C}^2) \oplus \mathbb{C}$.

9 Representations of $SU(3)$

We take the subgroup of $SU(3)$ of diagonal matrices, isomorphic to $U(1) \times U(1)$

$$T = \left\{ \begin{pmatrix} e^{i\theta_1} & 0 & 0 \\ 0 & e^{i\theta_2} & 0 \\ 0 & 0 & e^{-i(\theta_1+\theta_2)} \end{pmatrix} \mid (e^{i\theta_1}, e^{i\theta_2}) \in U(1) \times U(1) \right\}$$

Lemma 9.1. If $R|_T : SU(3) \rightarrow GL(V)$ is a complex representation, then $V = \bigoplus V[k, l]$, where

$$V[k, l] = \left\{ v : R \begin{pmatrix} e^{i\theta_1} & 0 & 0 \\ 0 & e^{i\theta_2} & 0 \\ 0 & 0 & e^{-i(\theta_1+\theta_2)} \end{pmatrix} v = e^{i(k\theta_1+l\theta_2)} v, \quad \forall \theta_1, \theta_2 \right\}$$

Proof. Let $T_1 = \{D(\theta_1, 0) = \begin{pmatrix} e^{i\theta_1} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{-\theta_1} \end{pmatrix}\} \cong U(1)$ and T_1 is a subgroup of T . So $V = \bigoplus U[k]$, with

$U[k] = \{v \in V : R(D(\theta_1, 0))v = e^{ik\theta_1}v\}$. Let $T_2 = \{D(\theta_1, 0) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & e^{i\theta_2} & 0 \\ 0 & 0 & e^{-i\theta_2} \end{pmatrix}\} \cong U(1)$. And note that if $v \in U[k]$, then $R(D(\theta_1, 0))v = e^{ik\theta_1}v$ and note that $D(\theta_1, 0)$ and $D(\theta_2, 0)$ are diagonal, they commute with one another, and hence the same is true for $R(D(\theta_1, 0)), R(D(\theta_2, 0))$. Therefore,

$$\begin{aligned} R(D(\theta_1, 0))R(D(\theta_2, 0))v &= R(D(0, \theta_2))R(D(\theta_1, 0))v \\ &= R(D(0, \theta_2))e^{ik\theta_1}v \\ &= e^{ik\theta_1}R(D(0, \theta_2))v \end{aligned}$$

Then $R(D(0, \theta_2))v \in U[k]$. This implies the lemma $U[k] = \bigoplus V[k, l]$, where $V[k, l] = \{v \in U[k] : R(D(0, \theta_2))v = e^{il\theta_2}v\}$. In other words, the elements of $V[k, l]$ transform as desired under both T_1 and T_2 and hence under T

$$\begin{aligned} R(D(\theta_1, \theta_2))v &= R(D(\theta_1, 0))R(D(0, \theta_2))v \\ &= e^{i(k\theta_1+l\theta_2)}v \end{aligned}$$

□

Example. Let $R = \mathbb{C}^3$ be the standard representation of $SU(3)$. We know $\mathbb{C}^3$ has a basis e_1, e_2, e_3 . We have

$$R \left(\begin{pmatrix} e^{i\theta_1} & 0 & 0 \\ 0 & e^{i\theta_2} & 0 \\ 0 & 0 & e^{-i(\theta_1+\theta_2)} \end{pmatrix} \right) = \left(\begin{pmatrix} e^{i\theta_1} & 0 & 0 \\ 0 & e^{i\theta_2} & 0 \\ 0 & 0 & e^{-i(\theta_1+\theta_2)} \end{pmatrix} \right)$$

Hence $e_1 \mapsto e^{i\theta_1}e_1, e_2 \mapsto e^{i\theta_2}e_2$ and $e_3 \mapsto e^{-i(\theta_1+\theta_2)}e_3$. So $V[1, 0] = \mathbb{C}e_1, V[0, 1] = \mathbb{C}e_2$ and $V[-1, -1] = \mathbb{C}e_3$.

The Lie algebra of $SU(3)$ is $su(3)$ which is 3×3 trace-free skew-Hermitian matrices. The complexification of $su(3)$ is given by $\mathfrak{sl}(3, \mathbb{C})$ trace-free 3×3 complex matrices. The basis of $\mathfrak{sl}(3, \mathbb{C})$ is given by E_{ij} and H_{ij} with

$$H_{13} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \text{ and } H_{23} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

More generally $H_\theta = \begin{pmatrix} \theta_1 & 0 & 0 \\ 0 & \theta_2 & 0 \\ 0 & 0 & \theta_3 \end{pmatrix} \in \mathfrak{sl}(3, \mathbb{C})$ such that $\theta_1 + \theta_2 + \theta_3 = 0$.

Example (Adjoint Representation). G is a Lie group and let

$$Ad : G \rightarrow GL(g)$$

$$Ad(g)X = gXg^{-1}$$

and $ad : g \rightarrow \mathfrak{gl}(g)$ with $ad(X)Y = [X, Y]$. Let's take $G = SU(3), g = su(3)$ then $ad^{\mathbb{C}} : \mathfrak{sl}(3, \mathbb{C}) \rightarrow \mathfrak{gl}(\mathfrak{sl}(3, \mathbb{C}))$. Then

$$ad(H_\theta)H_{ij} = [H_\theta, H_{ij}] = 0$$

since $H_\theta \rightarrow$ exponentiates to $D(\theta_1, \theta_2)$. Hence H_{ij} has weight 0. In addition,

$$\begin{aligned} ad(H_\theta)E_{ij} &= [H_\theta, E_{ij}] \\ &= (\theta_i - \theta_j)E_{ij} \end{aligned}$$

The weights of E_{ij} are $\theta_i - \theta_j$.

$$Ad(D(\theta_1, \theta_2))v = e^{i(k\theta_1+l\theta_2)}v$$

then $\exp(iad(H_\theta))v = e^{i(k\theta_1+l\theta_2)}v$ then

$$ad(H_\theta)v = k\theta_1 + l\theta_2$$

Thus, $ad(H_\theta E_{12}) = (\theta_1 - \theta_2)E_{12}, E_{12} \in V[1, -1]$ and $E_{13} \in V[2, 1], E_{32} \in V[-1, -2]$ etc.

We'll change notation for $V[n, l]$ satisfying $V[k, l] = V[\theta]$ where $\lambda(\theta) = k\theta_1 + l\theta_2$. So we are encoding k, l into the linear function λ . Some examples of linear functions

$$L_1(\theta) = \theta_1, L_2(\theta) = \theta_2, L_3(\theta) = -\theta_1 - \theta_2$$

These are the λ that correspond to $(k, l) = (1, 0), (0, 1), (-1, -1)$. L_1, L_2, L_3 are the weights of the standard representation $\mathbb{C}^3$ of $SU(3)$. The weights of the adjoint representation $L_i - L_j$. Recall that

$$ad(H_\theta)E_{ij} = (\theta_i - \theta_j)E_{ij} = (L_i - L_j)(\theta)E_{ij}$$

so $E_{ij} \in V[L_i - L_j]$.

Definition 9.2. The weight spaces of the adjoint representation are called root spaces, the weights that occur are called roots.

In the analogue of $SU(3)$ we need to check weight vectors $E_{ij} \in \mathfrak{sl}(3, \mathbb{C})$ of the adjoint representation act on the weight spaces of another representation.

Lemma 9.3. Given a Lie group representation $R : SU(3) \rightarrow GL(V)$ then $r^\mathbb{C}(E_{ij}) : V[\lambda] \rightarrow V[\lambda + L_i - L_j]$.

Proof. Suppose $v \in V[\lambda] := r^\mathbb{C}(H_\theta)v = \lambda(\theta)v$. We want to show $r^\mathbb{C}(E_{ij})v \in V[\lambda + L_i - L_j]$ i.e. $r^\mathbb{C}(H_\theta)r^\mathbb{C}(E_{ij})v = (\lambda + L_i - L_j)(\theta)r^\mathbb{C}(E_{ij})v$. Consider $r^\mathbb{C}(H_\theta)r^\mathbb{C}(E_{ij})v$,

$$\begin{aligned} r^\mathbb{C}[H_\theta, E_{ij}] &= [r^\mathbb{C}H_\theta, r^\mathbb{C}E_{ij}] \\ &= r^\mathbb{C}(H_\theta)r^\mathbb{C}(E_{ij}) - r^\mathbb{C}(E_{ij})r^\mathbb{C}(H_\theta) \end{aligned}$$

Thus,

$$\begin{aligned} r^\mathbb{C}(H_\theta)r^\mathbb{C}(E_{ij})v &= ((L_i - L_j)(\theta)r^\mathbb{C}(E_{ij}) + r^\mathbb{C}(E_{ij})r^\mathbb{C}(H_\theta))v \\ &= (L_i - L_j)(\theta)r^\mathbb{C}(E_{ij})v + \lambda(\theta)r^\mathbb{C}(E_{ij})v \\ &= (\lambda + L_i - L_j)r^\mathbb{C}(E_{ij})v \end{aligned}$$

□

Remark. We have

- (1) $r^\mathbb{C}$ is a Lie algebra representation, we can take it inside the bracket.
- (2) $ad(H_\theta)E_{ij} = [H_\theta, E_{ij}] = (\theta_1 - \theta_2)E_{ij} = (L_i - L_j)(\theta)E_{ij}$.

Lemma 9.4. $E_{ij}, E_{ji}, H_{ij} = [E_{ji}, E_{ij}]$ span a Lie subrepresentation of $\mathfrak{sl}(3, \mathbb{C})$ isomorphic to $\mathfrak{sl}(2, \mathbb{C})$.

Proof. We have 3 Lie algebra subrepresentations of $\mathfrak{sl}(3, \mathbb{C})$ isomorphic to $\mathfrak{sl}(2, \mathbb{C})$. Consider E_{12}, E_{21}, H_{12} where

$$E_{12} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, E_{21} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Moreover, we take Lie brackets and the rows/columns of O_3 (that we ignore) do not affect the computation i.e. E_{ij}, E_{ji}, H_{ij} satisfying the bracket relation of $X, Y, H \in \mathfrak{sl}(2, \mathbb{C})$. □

Corollary 9.5. The weight diagram of any complex $SU(3)$ representation has 3 lines of reflective symmetry.

Proof. This follows from the fact that the weight diagram of $\mathfrak{sl}(2, \mathbb{C})$ representation has a single line of reflective symmetry. □